

èfjç» - ç” µè`❖i¼š188-5665-3017 - é,®ç®±i¼šhs3434@foxmail.com -
ä, º°ç½ ‘ç«™i¼šhs3434.github.io - **æ±
,è❖Ææ,,❖â❖ “**i¼šAMD ROCm Python â❖Žç«`â¼æâ❖ ‘â • ¥ç`â, ^i¼^ ROCm Radeon Cloud
æ-¹â❖ ‘i¼% --- ## ä, º°ç®æ» 3 â¹ ‘ç” Ýç%©äzjæ❖-â • ¥ç`âÆ-ç»❖éâEi¼Æè ± ºâ|è½-âžç
K8s â❖Žç«`æ ^â€, \*\*â®¹â™™âÆ-éf`ç½²â€❖èµ,,æ°❖è°fâ°|â€❖ä»»âŠiç¼-æž’ â€❖GPU
æž`ç❖ + æœ❖âŠi\*\*â€, æœ€èz ‘2 â ‘`è ± â| ROCm / K8s / Helm / FastAPI /
Prometheusi¼Æçç-çç«`â®Ææ^❖ [**k8s-llm-runtime**](https://github.com/hs3434/k8s-llm-
runtime)i¼^K8s ä, Šçš,, vLLM æ`jâžçæœ❖âŠiç½ ‘â...³i¼%oi¼Æ**v0.1.0
taggedi¼Æâ❖-çž°âœ°è • ‘demo**â€, ç + Ýæ,% Snakemake ç±» Argo Workflows çš,, ML
â¹³â❖æš½è ± jâ€, --- ## æ±,è❖Ææš€èf½i¼^æ€%â²-â½❖âÆ¹é...❖æž’ â°❖i¼Æ4 â±
,i¼% ### â❖Žç«`æ , âz f - Python 3.11+i¼^asyncio, type hints, Pydantic, httpx asynci¼% -
FastAPIi¼^OpenAPI / dependency injection / lifespan¼% - æµè` • i¼špytest, pytest-asyncio,
88% coverage ### â®¹â™™`ä, Žç¼-æž’ - Dockeri¼^multi-stage æž,,â°i¼ÆAlpine
é • œâf❖ä¼~âÆ-i¼% - Kubernetesi¼^Pod / Service / Deployment / StatefulSet / ConfigMap /
RBACi¼% - Helm charti¼^values æ`jæ❖zâÆ-i¼Æhelm install/upgrade/uninstalli¼ÆCI lint
cleani¼% - é • œâf❖â» “â° “i¼šregistry.k8s.io / docker.io / ghcr.io / nvcr.ioi¼^â½â + ...
DaoCloud é • œâf❖âš é€Yi¼% ### GPU / LLM ç” Ýæ€❖ - vLLMi¼^OpenAI â...¼â®¹
APIi¼Ææž`ç❖ + æœ❖âŠi¼% - AMD ROCm âÝ°ç;€i¼^HSA / HIP æ|,âzµi¼Æamd.com/gpu
èµ,,æ°❖è°fâ°i¼% - PyTorch¼^CNN æ— ¶ ç©°è § £ç ❖â™™ / Transformeri¼% - Hugging
Facei¼^transformers â° “i¼Ætokenizeri¼% ### â❖-è § ,æµæ€ § ä, Žâ • ¥â... é “¾ -
Prometheusi¼^prometheus_clienti¼Æhistogram / counter / gaugei¼% - structlogi¼^JSON
structured loggingi¼% - uv / ruff / mypy strict / GitHub Actions - Linux
æœ❖âŠiç«`èzç»`i¼^systemd, Nginx, ç«`â❖£/èzç`çç°iç❖ + i¼% --- ## éiççç»❖éâÆ ###
k8s-llm-runtimei¼šK8s ä, Šçš,, vLLM æ`jâžçæœ❖âŠiç½ ‘â...³i¼^v0.1.0 taggedi¼% **é—
®èç~**i¼^â-¹â°” JD è❖Æè´£ 1 + 3i¼%oi¼švLLM æ`jâžçéf`ç½²éœ€ GPU + â®¹â™™ +
è • `ç” ±i¼Æâ»çé~Ýâšä°â...±â°æ— ¶ æ`jâžççç°iç❖ + â»æ❖, i¼ç” ç© ¶ /
â°❖â»çé~Ýæf³ç” “ OpenAI â...¼â®¹ API æ— ¶ é—`æ § é«~â€, i¼^ç»´æž¥â-¹â°”
JD i¼š”â» § æ`jâžçèç»f/æž`ç❖ + Workshopâ€❖æ • ™ç`â€❖âšÝèf½æ¼” ç»°”i¼%
æž ¶ æž,,i¼^â❖Æ chart + ä, %â±, lib + é • œâf❖ä¼~âÆ-i¼%oi¼š - `charts/llm-inference/
`i¼švLLM è´Yè½½ charti¼Æ\*\*GPU vendor â ± æ❖çâ»...é❖ `values.yaml` **i¼^amd.com/
gpu / nvidia.com/gpu / none ä, %æ€❖i¼Ævalues æ~¾¼¼❖é€%æç©i¼% - `charts/llm-router/
`i¼šFastAPI ç½ ‘â...³ chart - ConfigMap æ³`â...¥ initContainer âš è½½â...¾¼Ý chart - RBAC
æœ€â®¹❖æ°fè™™i¼^Pod/Service/ConfigMap read+write + Leasei¼% - ServiceMonitor â❖-é€
%â❖-ç” “i¼^Service 0.0.0.0:9090/metricsi¼% - `src/k8s\_llm\_runtime/`i¼š3 â±, lib æš½è ±
j - `K8sJobOperator`i¼^â½Žâ±,i¼Æéšç” “ Job è°fâ°i¼% - `VLLMInferenceOperator`i¼^â, -
â±,i¼Æhelm CLI subprocess + GPU vendori¼% - `ModelOperator`i¼^é«~â±,i¼ÆOpenAI â...
¼â®¹ + è ± âš`éf`ç½²i¼% **4 â» § æš€æœ-ä°ç,¹**i¼š 1. **lib ä, %â±,â + â±,æš½è ±
j**i¼^â-¹â°” JD â❖Žç«`æž ¶ æž,,èf½âš i¼% - ä, Šâ±, ModelOperator â❖â...¾âz f
load_model(alias) / unload_model(alias) - ä, â±, VLLMInferenceOperator â❖è£... helm
ç» + èš, - â½Žâ±, K8sJobOperator éšç” “ K8s Job è°fâ° 2. **Helm ConfigMap æ³`â...¥
aliases**i¼^â-¹â°” JD â®¹â™™âÆ- + èµ,,æ°❖è°fâ°i¼% - é❖zâ...❖æ-❖æ-jæ-°âçžæ`jâžçéf½
rebuild é • œâf❖i¼^GPU é • œâf❖ 15-20GB rebuild 5-10mini¼% - ConfigMap
ç § ‘ ç° § ç” Ýæ • ^i¼Æé ¶ é • œâf❖é ± ❖â° 3. **K8s Lease â + â, fâ¼❖é” ❖**i¼^â-¹â°”
JD é + ç¾æèzç»´ + HAi¼% - coordination.k8s.io/v1 Lease é²æç Router
â»šâ% -æœ-â¹ ¶ â❖ ‘éf`ç½²â❖Æä, €æ`jâžç-é” ❖ç²’ â°|æ€% model
aliasi¼Æè ± âš`èz ± æœÝ + çç° 4. **â, šâŠiç° § Prometheus metrics**i¼^â-¹â°” JD
â¹³â❖â❖-è § ,æµæ€ § i¼% - `chat\_completions\_duration\_seconds`i¼^histogrami¼Æâ❖«

status code æ þ ç³⁄⁴i¹⁄⁴% - `http\_requests\_total`i¹⁄⁴^counteri¹⁄⁴Ææ% method/route/statusi¹⁄⁴%
- `models\_loaded`i¹⁄⁴^gaugei¹⁄⁴Æâððæ~ in-memory statei¹⁄⁴% - ServiceMonitor âð - é€%è¿
Grafana **GPU vendor â^ þ æð¢ demo**i¹⁄⁴^â¹âº JD âš â^ þ GPU é¿ þ ç³⁄⁴m¹⁄⁴%oi¹⁄⁴š -
values.yaml `gpu.vendor=amd|nvidia|none` - AMDi¹⁄⁴šamd.com/gpu èµ,,æððè°fâºi¹⁄⁴ÆROCm
vLLM é • œâfð - NVIDIAi¹⁄⁴šnvidia.com/gpu èµ,,æððè°fâºi¹⁄⁴ÆCUDA vLLM é • œâfð -
nonei¹⁄⁴šCPU only æ`jâ¹⁄⁴ð¹⁄⁴^kind cluster é»~èºm¹⁄⁴Ææœ-âœ°â¹⁄⁴€âð ‘ç’ `i¹⁄⁴% **è
`é þ ðä¿ðè-ð**i¹⁄⁴š - 31 commits / 90 tests pass / coverage 88% - ruff + mypy strict + helm-lint
â...` clean - 2 Helm chartsi¹⁄⁴^llm-inference + llm-routeri¹⁄⁴% **è, ©âð ‘ä, Žä¿âðð**i¹⁄⁴š -
Service DNS ä, ðâÆ¹é...ð bugi¹⁄⁴š`--set fullnameOverride={release\_name}` ä¿âðð - kind
èš,ç,¹â¹⁄²â þ ...é • œâfðâš é€Yi¹⁄⁴šcontainerdConfigPatches + DaoCloud mirror -
è¿™äºèf¹⁄²â þ ¤â...¥ `docs/amd-interview-demo.md` ä³⁄⁴çŽ°âœ°âð,è€f
æœ°æð¥i¹⁄⁴^v1.1i¹⁄⁴%oi¹⁄⁴šSSE æµðâ¹⁄⁴ðæŽ`çð þ / RBAC æ” ¶ ç´ s âðè-» / çœYâºŽ GPU
é¿ þ ç³⁄⁴e2e ### âYºäºŽ Snakemake çš,, ML
â¹³âððâ • ¥â¹⁄²œæµðæj þ æž ¶i¹⁄⁴^ç” Yä¿jèfÆæ™`i¹⁄⁴Æç±» Argo Workflowsi¹⁄⁴% -
èfÆæ™`i¹⁄⁴šSnakemake rule è-æ³ • æ— æ³ • âºžçŽ°æ`jâð—ésðæ,,ðç,,âð^ -
èº³⁄⁴èºi¹⁄⁴šæ “â¹⁄⁴f rule è-æ³ • i¹⁄⁴Æç´ æŽ¥è°fâ þ ...éf`æŽ¥âð£âš`æ€ðç” Yæ^ð
rulei¹⁄⁴Æè s,,èÆfè³⁄⁴ “â...¥è³⁄⁴ “â þ °ç±»âžçâð³âð`ç s -æœ`â¹⁄⁴ðæç¹⁄⁴æŽ¥ - â...³èð” i¹⁄⁴šJD
âš â^ þ é¹¹æœ°â™`âjâ¹ â¹³âððç” âð ‘ç»ðè°Æ”â€” â€” æœ-éj¹çºæ~ ML â¹³âððæš¹⁄²è±
jçš,,â • ¥ç`âºžè • µ ### è,, ‘æœ°æŽ¥âð£ CNN è s £ç ðâ™`i¹⁄⁴^PyTorch¹⁄⁴%â€” â€” 6 â “â -
|ä¹ è • `çº¿äº s â þ ° - ç«-â^°ç«-æ- ¶ çº°âð • ç s -è s £ç ð EEG ä¿jâð • - PyQt âºžæ- ¶ GUI +
FastAPI æš´éœ² - è-|è s ð BCI è • `çº¿çº€âŽ þ [github.com/hs3434/learning-journey](https://
github.com/hs3434/learning-journey/blob/main/tracks/brain-computer-interface/notes/
resume.md) --- ## â • ¥â¹⁄²œç»ðâŽ þ ### ä, Šæµ • âºœ þ Æâðjâ¿jæð`ç s ‘æš€æœ%é™ðâ...
-âð, i¹⁄²œç” Yçº©â¿jæð`â • ¥ç`â, ^ i¹⁄²œ 2025.04 - è þ³âš - **Docker
éf`ç¹⁄²**i¹⁄⁴šä, šâšjâ»£ç ð Docker âÆ-i¹⁄⁴^âYºäºŽ snakemake + è þ³ç” æj þ æž ¶i¹⁄⁴% -
æœððâšjç«`è¿ðç»`i¹⁄⁴šLinux æœððâšjâ™`æ-¥â, ç»´ æšm¹⁄⁴ÆNginx
âðððâð ‘â»£çð þ i¹⁄⁴Æsystemd æœððâšjçºjçð þ - **âº ‘â¹³âððâ • ¥â...
• â¹⁄⁴€âð **i¹⁄⁴šä, šâšjè þ³âš`âÆ-æµðç`+ çº-æ³ • â • ¥â... • ###
ä, Šæµ • æ- s æ~ “ç” Yçº©â¿jæð`âjç s ‘æš€æœ%é™ðâ...-âð, i¹⁄²œ
ç” Yçº©â¿jæð`ç” âð ‘â • ¥ç`â, ^ i¹⁄²œ 2022.08 - 2024.07 - **âº ‘â¹³âððç” Yä¿jâ • ¥â...
• â¹⁄⁴€âð **i¹⁄⁴šR / Python / Linux / Docker / äº ‘âžYç” Y - **ä, šâšjè þ³âš`âÆ- **i¹⁄⁴šâÆ»â -
|è¹⁄²-â¹⁄² • ç»,,ä, šâšjæµðç`è þ³âš`âÆ-i¹⁄⁴Ææ`jâð—âÆ-â€ðâð`âððç” ` -
â¹³âððâœ°âð€i¹⁄⁴š[cloud.oebiotech.com](https://cloud.oebiotech.com/#/home) ###
ä, Šæµ • æ- s æ~ “ç” Yçº©â¿jæð`âjæœ%é™ðâ...-âð, i¹⁄²œç” Yçº©â¿jæð`âºžâ¹ ç” Y i¹⁄²œ
2021.10 - 2022.01 - âð • ç» þ èfžè¹⁄²-â¹⁄² • ç»,,i¹⁄⁴šè
`æž s æ€ðâ, è s,,â^ þ æžðæ€ðâ” âðžæ€ðä, æ€ s âÆ-â^ þ æžð -
ä, »è|ðæš€æœ-i¹⁄⁴šâð • ç» þ èfžè¹⁄²-â¹⁄² • ç»,,â€ðRâ€ðPythonâ€ðLinux --- ##
æ • ¤è,²èfÆæ™` **è¥¿jâÆ—â þ œæž—ç s ‘æš€âð s âj** i¹⁄²œ æðçº©ç s ‘â -
|ä, Žæš€æœ-ä, “ä, š i¹⁄²œ æœ-ç s ‘i¹⁄²œ 2018.09 - 2022.07 i¹⁄²œ GPAi¹⁄⁴š3.35 - æ- • ä, šèºæ-
þ i¹⁄⁴šâ€šâYºäºŽé€»è³⁄⁴ ‘æ-è’ ,â¿žâ¹⁄² çjâºšâºè°jâ€” ä¹⁄⁴ðç, â...³æ€ s çš ¶ âðšâš—
â€” æ€ s çš ¶ æÆ þ æ þ æ€ - âð s â^éj¹çºi¹⁄⁴šæ¹çºè°ÆæŽ¥ç s ðâ¹çº «èš±èœè “¿æš—
é “œæ±jæY “çš,,â¹⁄²±â “ð¹⁄⁴^âð,ä, Žâ¹⁄²œè€...i¹⁄⁴% --- ## ä, äººâ¹⁄⁴âš¿ - **6 â “è þ³âj K8s
âðžç«-æ`\*\*i¹⁄⁴šä»Žé¶ â^º v0.1.0 tagged éj¹çºi¹⁄⁴Æç«-ç«âºÆæ`âðÆ Helm chart + Python
lib - **çŽ°âœ° demo èf¹⁄²âš»**i¹⁄⁴šâð`è þ³â, |ç” µè,, ‘âœ` kind cluster ä, šè • ‘éšâºÆæ • ´
LLM æŽ`çð þ æµðç`i¹⁄⁴^éf`ç¹⁄²â þ’ è • `ç” ± â þ’ chat completions â þ’ metricsi¹⁄⁴% -
âjâ¹ èf¹⁄²âš»â¹⁄⁴ºi¹⁄⁴š3 â¹`â þ...â»Žç” Yçº©â¿jæð`è • `ç • Æâº K8s
âðžç«-i¹⁄⁴Ææš€æœ-çfæf...é«- **3 â¹` Python â • ¥ç`âÆ-
ç»ðè°Æ**i¹⁄⁴šä»Žç s ‘ç” è,,šæœ-â^ºâº ‘â¹³âððâ • ¥â... • çš,,âºÆæ • ´è • -â³⁄⁴,,